Question ID: 184ce5aa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question
Line h is defined by $\frac{1}{5}x+\frac{1}{7}y-70=0$. Line j is perpendicular to line h in the *xy*-plane. What is the slope of line j?

Answer
A. $-\frac{7}{5}$

B. $-\frac{5}{7}$

C. $\frac{7}{5}$

D. $\frac{5}{7}$

Correct Answer: D

Rationale
Choice D is correct. It's given that line h is defined by $\frac{1}{5}x+\frac{1}{7}y-70=0$. This equation can be written in slope-intercept form $y=mx+b$, where m is the slope of line h and b is the *y*-coordinate of the *y*-intercept of line h. Adding 70 to both sides of $\frac{1}{5}x+\frac{1}{7}y-70=0$ yields $\frac{1}{5}x+\frac{1}{7}y=70$. Subtracting $\frac{1}{5}x$ from both sides of this equation yields $\frac{1}{7}y=-\frac{1}{5}x+70$. Multiplying both sides of this equation by 7 yields $y=-\frac{7}{5}x+490$. Therefore, the slope of line h is $-\frac{7}{5}$. It's given that line j is perpendicular to line h in the *xy*-plane. Two lines are perpendicular if their slopes are negative reciprocals, meaning that the slope of the first line is equal to -1 divided by the slope of the second line. Therefore, the slope of line j is the negative reciprocal of the slope of line h. The negative reciprocal of $-\frac{7}{5}$ is $\frac{-1}{\left(-\frac{7}{5}\right)}$, or $\frac{5}{7}$. Therefore, the slope of line j is $\frac{5}{7}$.

Choice A is incorrect. This is the slope of a line in the *xy*-plane that is parallel, not perpendicular, to line h.

Choice B is incorrect. This is the reciprocal, not the negative reciprocal, of $-\frac{7}{5}$.

Choice C is incorrect. This is the negative, not the negative reciprocal, of $-\frac{7}{5}$.